THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · HARD LEVEL UNIT TEST SET 1

Chapters 1–4: **The Wonderful World of Science, Diversity in the Living World, Mindful Eating: A Path to a Healthy Body, Exploring Magnets** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. To find whether the volume of water affects its boiling time, a student boils 100 mL in a steel pot on a strong flame and 200 mL in a glass beaker on a weak flame. Why is this test unfair? 1

(a) the container and the flame also differ along with the volume
(b) the water is heated at all
(c) time is measured with a clock
(d) the volumes of water are different

2. A scientist finds a plant with parallel veins, fibrous roots and a soft stem. Which is most probable? 1

(a) a grass **(b)** a peepal tree **(c)** a mango tree **(d)** a rose bush

3. A food gives a blue-black colour with iodine and an oily patch on paper. It contains: 1

(a) protein and fibre **(b)** starch only
(c) fat only **(d)** starch and fat

4. Three identical-looking bars P, Q, R: P attracts Q at both ends, Q attracts R at both ends, R repels P at one end. Which can be concluded? 1

(a) P and R are magnets, Q is magnetic material
(b) all are plastic
(c) all are iron
(d) all are magnets

5. A study found that more ice cream is sold when more people get sunburnt. The best explanation is: 1

(a) both depend on hot weather **(b)** sunburn causes ice cream sales
(c) ice cream causes sunburn **(d)** it is magic

6. Which of the following is a correct inference about a tiger's eyes in front of the face? 1

(a) helps see all sides **(b)** helps fly
(c) helps judge distance for hunting **(d)** helps to swim

7. Why do nutritionists advise limiting packaged drinks? 1

(a) they have fibre
(b) they contain much sugar and few nutrients
(c) they contain protein
(d) they have vitamins

8. Which of the following is the best argument for using a magnet to separate iron from copper turnings? 1

(a) iron is magnetic and copper is not **(b)** both are shiny
(c) both are metals **(d)** both rust

9. Suppose two groups test the same hypothesis and get opposite results. What should be done first? 1

(a) believe one group **(b)** toss a coin
(c) ignore both **(d)** compare the methods and repeat carefully

10. In a forest, if all insects vanish, what is most likely to happen? 1

(a) forest will grow faster **(b)** all plants will grow faster
(c) many plants will fail to pollinate **(d)** animals will grow bigger

11. A paper rubbed with a food sample shows a translucent oily patch that stays even after drying. The food contains: 1

(a) starch **(b)** fat **(c)** sugar **(d)** fibre

12. To make a magnet from an iron bar by the single-touch method, the bar should be stroked: 1

(a) always in the same direction with the same pole of a magnet
(b) back and forth with both poles
(c) by hitting it with the magnet
(d) in a circle with the magnet

13. **Assertion (A):** A hypothesis is a guess that can be tested. 1
Reason (R): An experiment can show whether a hypothesis is supported or not.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. **Assertion (A):** Wheat plants have fibrous roots. 1
Reason (R): Plants with fibrous roots generally have leaves with parallel venation.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. **Assertion (A):** Proteins are called body-building nutrients. 1
Reason (R): Proteins help in growth and repair of the body.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. In an experiment on plant growth, A is kept in sunlight and B in shade, but A is also given more water. Why is this unfair? 2

17. Cactus leaves are spines but it still makes food. How? 2

18. A food sample turns blue-black with iodine and leaves an oily spot on paper. What conclusions can you draw? 2

19. A compass is placed on a steel table and gives a wrong reading. Explain and suggest a remedy. 2

20. How can a scientific model such as a globe help us? 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Design a fair test to compare which of three fabrics dries the fastest. Name controlled variables. 3

22. A plant has leaves with net veins, a tap root, a thick woody trunk 15 m tall. Classify it and give the chief features of each part. 3

23. A girl's diet: breakfast - bread and jam, lunch - noodles, dinner - fried snacks. Evaluate it and suggest changes. 3

24. Describe how to show that the poles of a magnet are not at the very ends but near them, using iron filings. 3

OR

Explain the role of curiosity, observation and reasoning in the discovery of gravity by Newton. 3

25. A student suggests classifying animals by colour. Why is this not a good method? 3

OR

Why should dairy foods be included in a school child's diet? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** Riya noticed that the plant on her balcony grew taller than the plant kept inside her cupboard. She guessed that plants need sunlight to grow well. She took two identical pots with the same soil and gave both the same amount of water. She kept one in sunlight and the other in a dark cupboard, and noted the height of both plants every two days for ten days. 5

(i) What is Riya's guess called?

(ii) Which one thing was different for the two pots?

(iii) Name the step in which she noted the heights every two days.

(iv) Why did she keep the soil and water the same? What is the likely conclusion?

27. A group of students are tested for health. Student A has bleeding gums, B has night blindness, C has swollen neck, D has bent legs, E is pale. Identify the cause, the remedy for each. 5

OR

Design an investigation to compare the strength of two magnets using paper clips and describe how to make it a fair test. 5

END OF QUESTION PAPER — Total: 50 Marks